Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

Learning goal: I can use order of operations with rational numbers. (circle your level after scoring)

4	I can find and explain an error in someone else's work with order of operations.
3	I can evaluate an expression with rational numbers, exponents, and grouping symbols.
2	I can use order of operations with whole numbers.
1	I need help to start this benchmark.

1. Evaluate the expression.

$4 \times (|4.57 - 8.3| - 0.8^2)$

2. What is the value of the expression?

$7.3 - [0.75 + (-0.5 - 2)^2] \div 0.5$

A. 0.60

B. −6.70

C. 19.05

D. 18.30

Turn over →

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____

3. Evaluate the expression. Write your answer as a fraction in simplest form.

$$9 - 2[\left(1\frac{1}{2} - 2\right)^2 + \frac{1}{2} \div 6]$$

4. What is the value of the expression? Write your answer as a fraction in simplest form.

$$[\left(2\frac{1}{2} - 3\right)^2 + |\frac{3}{8} - 4|] \times 3$$

5. Where should one pair of parentheses be placed so that the expression has a value of 147.75?

$$-8.5 + 3 \times 5 - 2.5^2$$

A. $-8.5 + 3 \times (5 - 2.5^2)$

B. $-8.5 + 3 \times (5 - 2.5)^2$

C. $-8.5 + (3 \times 5 - 2.5)^2$

D. $(-8.5 + 3) \times 5 - 2.5^2$

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.2.1 · MA.7.NSO.2.1 · Order of operations with rational numbers

Unit 8: Evaluating Numeric Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **12.36**

   Inside the absolute value: $4.57 - 8.3 = -3.73$, so $|4.57 - 8.3| = 3.73$.
   The exponent: $0.8^2 = 0.64$.
   Inside the parentheses: $3.73 - 0.64 = 3.09$.
   Multiply: $4 \times 3.09 = 12.36$.

2. Regular **B. −6.70**

   Innermost grouping first: $-0.5 - 2 = -2.5$.
   The exponent: $(-2.5)^2 = 6.25$.
   Inside the brackets: $0.75 + 6.25 = 7$.
   Divide: $7 \div 0.5 = 14$.
   Subtract: $7.3 - 14 = -6.7$.
   Subtracting before dividing gives 0.60; dropping the brackets gives 19.05.

3. Regular $8\frac{1}{3} = \frac{25}{3}$

   Inside the parentheses: $1\frac{1}{2} - 2 = -\frac{1}{2}$.
   The exponent: $\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$.
   Divide: $\frac{1}{2} \div 6 = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.
   Inside the brackets: $\frac{1}{4} \div \frac{1}{12} = \frac{1}{3}$.
   Multiply: $2 \times \frac{1}{3} = \frac{2}{3}$.
   Subtract: $9 - \frac{2}{3} = 8\frac{1}{3}$.

4. Regular $11\frac{5}{8} = \frac{93}{8}$

   Inside the parentheses: $2\frac{1}{2} - 3 = -\frac{1}{2}$.
   The exponent: $\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$.
   Inside the absolute value: $\frac{3}{8} - 4 = -3\frac{5}{8}$, so $\left|\frac{3}{8} - 4\right| = 3\frac{5}{8}$.
   Inside the brackets: $\frac{1}{4} + 3\frac{5}{8} = 3\frac{7}{8}$.
   Multiply: $3\frac{7}{8} \times 3 = 11\frac{5}{8}$.

5. Rigor **C.** $-8.5 + (3 \times 5 - 2.5)^2$

   Evaluate each choice.
   $(-8.5 + 3) \times 5 - 2.5^2 = -33.75$
   $-8.5 + 3 \times (5 - 2.5)^2 = 10.25$
   $-8.5 + (3 \times 5 - 2.5)^2 = 147.75$
   $-8.5 + 3 \times (5 - 2.5^2) = -12.25$
   Without parentheses the expression is 0.25. Only $-8.5 + (3 \times 5 - 2.5)^2$ has the value 147.75.

Re-check strip (3 items, C-NSO.2.1-R)

1. Regular $4\frac{5}{8} = \frac{37}{8}$

   Inside the parentheses: $1\frac{1}{4} - 2 = -\frac{3}{4}$.
   The exponent: $\left(-\frac{3}{4}\right)^2 = \frac{9}{16}$.
   Divide: $\frac{1}{4} \div 2 = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$.
   Inside the brackets: $\frac{9}{16} + \frac{1}{8} = \frac{11}{16}$.
   Multiply: $2 \times \frac{11}{16} = 1\frac{3}{8}$.
   Subtract: $6 - 1\frac{3}{8} = 4\frac{5}{8}$.

2. Regular **B. −16.30**

   Innermost grouping first: $1.5 - 3 = -1.5$.
   The exponent: $(-1.5)^2 = 2.25$.
   Inside the brackets: $3.75 + 2.25 = 6$.
   Divide: $6 \div 0.5 = 12$.
   Subtract: $-4.3 - 12 = -16.3$.
   Subtracting before dividing gives −20.60; dropping the brackets gives −3.55.

3. Regular $2\frac{5}{8} = \frac{21}{8}$ **(or 2.625)**

   Write the decimal as a fraction: $1.75 = 1\frac{3}{4}$.
   Inside the parentheses: $-2\frac{1}{2} - 1 = -3\frac{1}{2}$.
   The exponent: $\left(-3\frac{1}{2}\right)^2 = 12\frac{1}{4}$. A negative number squared is positive.
   Inside the brackets: $12\frac{1}{4} - 1\frac{3}{4} = 10\frac{1}{2}$.
   Divide: $10\frac{1}{2} \div 4 = 2\frac{5}{8}$.

Comparison strip (2 items, C-NSO.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **31.25**

   Inside the absolute value: $4.6 - 7.8 = -3.2$, so $|4.6 - 7.8| = 3.2$.
   Inside the parentheses: $5.7 - 3.2 = 2.5$.
   The exponent: $(2.5)^3 = 15.625$.
   Divide: $15.625 \div 0.5 = 31.25$.

2. Regular $2\frac{1}{4} = \frac{9}{4}$ **(or 2.25)**

   Write the decimal as a fraction: $1.75 = 1\frac{3}{4}$.
   Inside the parentheses: $-1\frac{1}{2} - 1 = -2\frac{1}{2}$.
   The exponent: $\left(-2\frac{1}{2}\right)^2 = 6\frac{1}{4}$. A negative number squared is positive.
   Inside the brackets: $6\frac{1}{4} - 1\frac{3}{4} = 4\frac{1}{2}$.
   Divide: $4\frac{1}{2} \div 2 = 2\frac{1}{4}$.

Reteach plan — C-NSO.2.1

MA.7.NSO.2.1 · Order of operations with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.2.1-R

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a fraction in simplest form.

$$6 - 2[\left(1\frac{1}{4} - 2\right)^2 + \frac{1}{4} \div 2]$$

2. What is the value of the expression?

$$-4.3 - [3.75 + (1.5 - 3)^2] \div 0.5$$

A. −7.30 **B.** −16.30

C. −3.55 **D.** −20.60

3. Evaluate the expression.

$$[\left(-2\frac{1}{2} - 1\right)^2 - 1.75] \div 4$$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.2.1-R

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a fraction in simplest form.

$$6 - 2[\left(1\frac{1}{4} - 2\right)^2 + \frac{1}{4} \div 2]$$

2. What is the value of the expression?

$$-4.3 - [3.75 + (1.5 - 3)^2] \div 0.5$$

A. −7.30 **B.** −16.30

C. −3.55 **D.** −20.60

3. Evaluate the expression.

$$[\left(-2\frac{1}{2} - 1\right)^2 - 1.75] \div 4$$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.1-X

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a decimal.

$(5.7 - |4.6 - 7.8|)^3 \div 0.5$

2. Evaluate the expression.

$[\left(-1\frac{1}{2} - 1\right)^2 - 1.75] \div 2$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.1-X

Benchmark: **MA.7.NSO.2.1** · Order of operations with rational numbers Student code: _____ Calculator: no

1. Evaluate the expression. Write your answer as a decimal.

$(5.7 - |4.6 - 7.8|)^3 \div 0.5$

2. Evaluate the expression.

$[\left(-1\frac{1}{2} - 1\right)^2 - 1.75] \div 2$

Score: ____ / 2 → **copy to your progress strip**